

Practice Program-3

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Q1. Define a function overlapping () that takes two lists and returns True if they have at least one member in common, False otherwise.

Program:-

```
lst1=input('Enter First List: ').split(' ')
lst2=input('Enter Second List: ').split(' ')

def overlapping(list1,list2):
    for i in range(0,len(list1)):
        for j in range(0,len(list2)):
            if list1[i]==list2[j]:
                return True
    return False
print(lst1)
print(lst2)
print(overlapping(lst1,lst2))
```

Q2.In English, present participle is formed by adding suffix -ing to infinite form: go -> going. A simple set of rules can be given as follows:

- a. If the verb ends in e, drop the e and add ing
- b. If the verb ends in ie, change ie to y and add ing

Write a function make_ing_form() which accepts a list of verbs and returns a dictionary with verb : present participle

Program:-

```
list =input('Enter the list: ').split(' ')
dict ={}
def make_ing_form(list):
    for verb in list:
        if verb.endswith('ie'):
            dict[verb] = verb[0:len(verb)-2]+'y'+ 'ing'
```

```

        elif verb.endswith('e'):
            dict[verb] = verb[0:len(verb)-1]+'ing'
        else:
            dict[verb]=verb+'ing'

make_ing_form(list)
print(dict)

```

Q.3. Function display_greeting(message) prints message sent as argument as today's greeting. Decorate the function using appropriate decorated so that the greeting is displayed using Uppercase.

Program:-

```

greet=input('Enter the greet: ')

def uppercase(func):
    def inner():
        return func(greet).upper()
    return inner

@uppercase
def display_greeting(greet):
    return greet

print(display_greeting())

```

Q. 4. Create series of 'n' prime numbers and display first 10

Program:-

```

n = int(input('Enter the number of prime numbers: '))

def prime_generator(n):
    count = 0
    num = 2
    while count < n:
        flag = True
        for i in range(2, num):

```

```

        if num % i == 0:
            flag = False
            break
    if flag:
        yield num
        count += 1
    num += 1

gen = prime_generator(n)
for i in range(n):
    print(next(gen))

```

Q.5

1. Find employees that know 'python'
 2. Add a new skill - 'test' in skillset of all employees
 3. Sort employees by skills
- for the given dictionary of employees

```
emp_data = {'Amol': ['C', 'C++', 'Java'], 'Aditya': ['Angular', 'Java'],
            'Aditi': ['Python', 'PHP', 'Database']}
```

Program:-

```

emp_data = {'Amol': ['C', 'C++', 'Java'],
            'Aditya': ['Angular', 'Java'],
            'Aditi': ['Python', 'PHP', 'Database']}

know_python=list(filter(lambda key : 'Python' in
emp_data[key],emp_data))
print(know_python)

# updated_dict=map(lambda key :
emp_data[key].append('Test'),emp_data.keys())
# print(next(updated_dict))
# print(emp_data)

updated_dict =list(map(lambda item:{item[0]:[item[1] +
['Test']]},emp_data.items()))

```

```

print(updated_dict)
print(emp_data)

sorted_dict = dict(sorted(emp_data.items(), key=lambda x:
x[1]))
print(sorted_dict)

```

Q.6

Following data displays min/max/average temp for cities

weather= [{'Mumbai': [28, 30, 32]},.....]

1. Print the weather data
2. Print the city with maximum/min temp
3. Print all the cities that experience min temp more than 30 degree
4. Create a dictionary to print 'City': 'Ave temp'

Program:-

```

weather = [{'Mumbai': [28, 30, 43]},
            {'Pune': [25, 29, 31]},
            {'Delhi': [20, 33, 38]},
            {'Chennai': [31, 33, 36]}]

# convert list of dictionaries into one dictionary
weather_dict = {city: temps for d in weather for city, temps
in d.items()}

# 1. Print weather data
print("Weather Data:")
print(weather_dict)

# 2. City with maximum and minimum temperature
max_city = max(weather_dict.items(), key=lambda x: max(x[1]))
min_city = min(weather_dict.items(), key=lambda x: min(x[1]))
print("City with Maximum Temperature:", max_city[0],
max(max_city[1]))
print("City with Minimum Temperature:", min_city[0],
min(min_city[1]))

```

```
# 3. Cities with minimum temp more than 30
cities = list(filter(lambda x: min(x[1]) > 30,
weather_dict.items()))
print("Cities with min temp > 30:", list(map(lambda x: x[0],
cities)))

# 4. Dictionary City : Average Temperature
avg_temp = {city: round(sum temps)/len(temps),2) for city,
temps in weather_dict.items()}
print("City Average Temperature:", avg_temp)
```

Outputs:

Q1.

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day4\
Enter First List: Python Programming
Enter Second List: Java Programming
['Python', 'Programming']
['Java', 'Programming']
True
```

Q2.

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day4\
Enter the list: time tie sign
{'time': 'timing', 'tie': 'tying', 'sign': 'signing'}

Process finished with exit code 0
```

Q3.

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Da
Enter the greet: Good Morning
GOOD MORNING

Process finished with exit code 0
|
```

Q4.

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day4\
Enter the number of prime numbers: 10
2
3
5
7
11
13
17
19
23
29
```

Q5.

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day4\Assignment\Q5.py
['Aditi']
[{'Amol': [['C', 'C++', 'Java', 'Test']], {'Aditya': [['Angular', 'Java', 'Test']], {'Aditi': [['Python', 'PHP', 'Database', 'Test']}]
{'Amol': ['C', 'C++', 'Java'], 'Aditya': ['Angular', 'Java'], 'Aditi': ['Python', 'PHP', 'Database']}
{'Aditya': ['Angular', 'Java'], 'Amol': ['C', 'C++', 'Java'], 'Aditi': ['Python', 'PHP', 'Database']}
```

Process finished with exit code 0

Q6.

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day4\Assignment\Q6.py
Weather Data:
{'Mumbai': [28, 30, 43], 'Pune': [25, 29, 31], 'Delhi': [20, 33, 38], 'Chennai': [31, 33, 36]}
City with Maximum Temperature: Mumbai 43
City with Minimum Temperature: Delhi 20
Cities with min temp > 30: ['Chennai']
City Average Temperature: {'Mumbai': 33.67, 'Pune': 28.33, 'Delhi': 30.33, 'Chennai': 33.33}

Process finished with exit code 0
```